
Tools, Not Talk: The Knowledge Ceiling of Biology Agents

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Abstract

1 Agentic systems for biology are increasingly built around orchestration: critics
2 that review answers, teams of role-playing scientists that debate, and models that
3 reason for longer. We show that on objectively scored biology research tasks this
4 machinery adds little beyond what matched independent sampling provides, and
5 we identify why. Across 150 pre-registered LAB-Bench and LABBench2 ques-
6 tions, compared at a matched budget of three model calls, self-critique never beats
7 simple majority voting, multi-agent debate helps only a model whose reasoning
8 is switched off, and a 25-fold increase in generated tokens from longer reasoning
9 does not improve accuracy. The reason is a ceiling set by the questions them-
10 selves: which question is asked explains 56% of the variance in correctness, while
11 the choice of model, strategy and reasoning effort explains 0.6%. That ceiling is
12 largely missing knowledge rather than missing deliberation. A single model call
13 with web search and code execution solves 47% of the questions that all eight
14 orchestrated systems in our main study failed. For life-science agents, the budget
15 belongs in tools, not talk.

16 1 Introduction

17 LLM agents now plan experiments, analyse data and generate hypotheses in chemistry and biology
18 [1, 2, 5, 6, 8, 16]. Many of these systems are organised as conversations: a critic reviews a draft,
19 several specialised scientists debate, an adjudicator decides. Their gains are usually reported against
20 a single model call, even though the agent makes many. Sampling several answers and voting already
21 improves accuracy [3, 12, 19], so these comparisons cannot tell whether the orchestration helps or
22 whether the extra calls do [9].

23 We separate the two. Every orchestrated strategy receives exactly the budget of a voting baseline:
24 three calls, the same model, the same prompts and the same decoding. We then ask what actually
25 moves accuracy on biology research tasks: orchestration, longer reasoning, or access to knowledge
26 through tools.

27 Contributions.

- 28 • **Orchestration adds little at matched compute.** Across three models, self-critique never
29 beats voting and breaks at least as many answers as it repairs. Debate helps only when the
30 model cannot reason on its own, and the gain disappears when the same model is allowed
31 to think. Raising reasoning effort, for six model families, rarely helps either (§3.1–3.2).
- 32 • **The question decides.** Systems largely solve the same questions and fail the same ques-
33 tions. Question identity explains 56% of the variance in correctness and the system 0.6%
34 (§3.3).

Table 1: Accuracy (%) at a matched budget of three calls (DIRECT uses one). Δ = DEBATE – VOTE in points, with paired bootstrap 95% CI.

Model	n	DIRECT	VOTE	CRITIQUE	DEBATE	Δ [95% CI]
GPT-OSS-120B	150	46.7	49.3	42.7	47.3	-2.0 [-8.7, +4.7]
Gemma 4 31B (no thinking)	150	48.7	48.7	47.3	57.3	+8.7 [+2.0, +15.3]
Flash-Lite	75	61.3	62.7	56.0	57.3	-5.3 [-13.3, +1.3]

- **Tools break the ceiling.** One tool-using call solves 47% of the questions that no main-study system solved, and more compute on top of tools adds comparatively little (§3.4).

2 Setup

Tasks. We draw 30 questions from each of five categories (150 in total): four from LAB-Bench [10] (protocol troubleshooting, sequence manipulation, database and literature questions; multiple choice) and open-ended sequence computations from LABBench2 SeqQA2 [11], scored with the official validators. All answers are scored deterministically, with no LLM judge.

Strategies. DIRECT makes one call. VOTE makes three independent calls and takes the majority answer. CRITIQUE solves, critiques its own solution, then revises. DEBATE has a molecular biologist and a skeptical reviewer solve independently, then an adjudicator decides. VOTE, CRITIQUE and DEBATE each use exactly three calls, and only visible outputs pass between calls (prompts in App. A).

Models. The matched-budget study uses GPT-OSS-120B [13] (low reasoning effort), Gemma 4 31B [17] (thinking disabled) and Gemini 3.5 Flash-Lite (low thinking) on all 150 questions (Flash-Lite on a fixed 75). To test reasoning effort and tools, we add GPT-6 Luna, GPT-6 Sol and GPT-5.6 Terra, and rerun GPT-OSS and Flash-Lite at high effort, on a fixed 30-question subset with 6 per category. Task IDs, prompts, decoding and the primary analysis were frozen before evaluation (App. B). All contrasts are paired over questions, with bootstrap 95% CIs and exact McNemar tests.

3 Results

3.1 Orchestration adds little at matched compute

Three calls spent on orchestration generally do no better than three calls spent on independent samples (Table 1). **Self-critique does not beat voting** for any of the three models (-6.7 , -1.3 , -6.7 points; pooled -4.5 $[-9.3, +0.3]$). It breaks correct answers as readily as it fixes wrong ones: relative to the vote it rescues 24%, 13% and 25% of errors but corrupts 38%, 16% and 26% of correct answers. The damage is worst on exact sequence manipulation (SeqQA: -23 and -20 points for GPT-OSS and Flash-Lite), where a fluent but wrong critique argues the model out of a correct computation. **Debate helps only the model that cannot think.** For Gemma, whose thinking is disabled, debate beats voting by +8.7 points ($p = 0.015$); for the two models that reason, it does not (-2.0 and -5.3).

3.2 Longer reasoning does not substitute for knowledge

For most models, raising reasoning effort multiplies the cost without moving accuracy (Fig. 1). GPT-OSS at high effort generates $25\times$ more tokens and scores 47% against 50%; Flash-Lite spends $5\times$ more tokens and scores 67% against 70%. The one model that gains is GPT-6 Luna, which uses no reasoning tokens at low effort, and its high-effort single call reaches exactly the accuracy its low-effort debate already had (60%). The same substitution appears within a single model. On the same 28 questions, Gemma gains +7 points from debate over voting with thinking disabled but -4 with thinking enabled, and one thinking call (75%) matches debate without thinking (75%) at $\sim 5,045$ generated tokens. Debate and reasoning are therefore substitutes, and neither gets past the

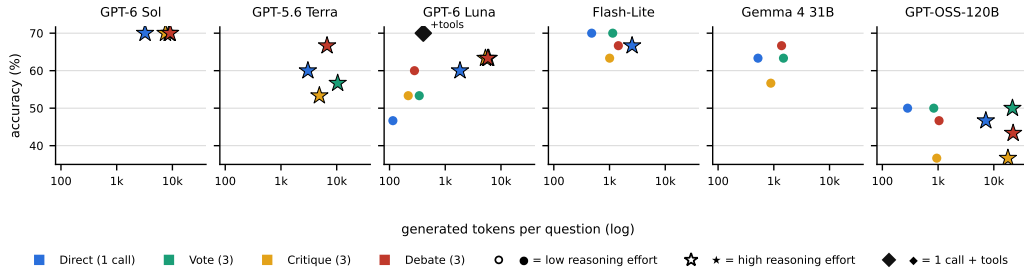


Figure 1: Every way of spending inference lands on a similar per-model plateau; tools rise above it. Accuracy vs. generated tokens per question on a shared 30-question set, one panel per model. Colour marks the strategy. Marker shape marks reasoning effort: circles (●) are low effort and stars (★) high effort. GPT-6 Sol and GPT-5.6 Terra were run at high effort only. Across nearly two orders of magnitude in tokens, accuracy moves little within a model. The black diamond (◆) is one GPT-6 Luna call with web search and code execution. It is the only configuration clearly above its model’s plateau, and among the cheapest.

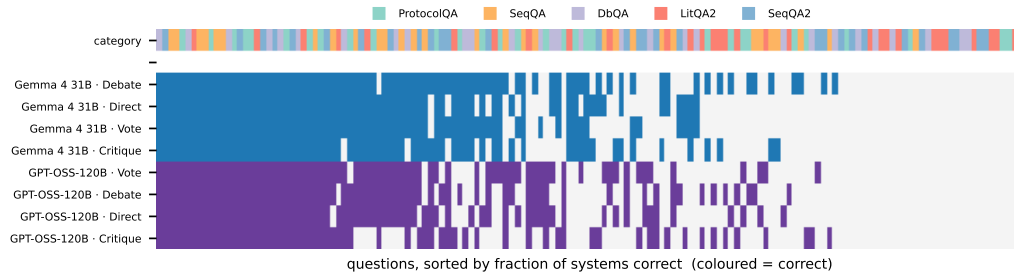


Figure 2: The question, far more than the system, decides. Correctness of every model × strategy system (rows) on every question (columns, sorted by the fraction of systems correct). Systems largely solve and fail the same questions. Question identity explains 56% [49, 62] of the variance in correctness; the system explains 0.6% [0.3, 1.9].

74 plateau. Stacking orchestration on high effort adds little: GPT-6 Sol scores 70% with every strategy
75 while the three-call strategies generate $\sim 3\times$ the tokens.

76 3.3 The question, not the system, decides

77 A two-way decomposition of the correctness matrix attributes 56% of the variance to the question
78 and 0.6% to the system (Fig. 2). The pattern holds across all 33 model, strategy and effort configura-
79 tions on the shared subset, frontier models at high effort included (47% vs. 3.9%). Systems largely
80 solve the same easy questions and fail the same hard ones. The 32 questions that no main-study
81 system solves are dominated by literature and database recall (18) and open-ended sequence com-
82 putation (9). These are failures of knowledge, which additional deliberation did not supply in any
83 configuration we tested.

84 3.4 Tools break the ceiling

85 Giving GPT-6 Luna web search and code execution in a single call raises accuracy on the shared
86 subset from 47% to 70% (+23 points [+7, +43], $p = 0.039$). That is more than any orchestration
87 or reasoning setting achieves, at ~ 404 generated tokens per question. On the 32 questions that all
88 eight main-study systems (two models × four strategies) failed, the same single call reaches 47%,
89 against 6% without tools (+41 points [+25, +56], $p = 0.0002$). Literature questions rise from 18%
90 to 91% and database questions from 0% to 43%. Retrieval helping on these categories is expected,
91 since they were designed for retrieval agents [10, 14]. The point is the contrast: none of the eight
92 orchestrated systems solved any of them. Compute on top of tools adds comparatively little. High
93 effort reaches 83% (+13, $p = 0.22$) and a three-call vote 77% (+7, $p = 0.73$), at $\sim 6\times$ and $\sim 3.5\times$
94 the tokens. Tool calls never reached the benchmark or its datasets.

95 4 Implications for building life-science agents

96 **Benchmark against a matched vote.** A multi-call agent should beat independent sampling with the
97 same number of calls, not a single call [9]. Most orchestration gains disappear under this baseline,
98 consistent with compute-matched studies of debate in general reasoning [4, 15, 18, 20] and with the
99 unreliability of intrinsic self-correction [7]. **Spend the budget on tools, not talk.** Critics, debaters
100 and longer chains of thought re-deliberate over knowledge the model does not have. Retrieval sup-
101 plies it, at a fraction of the tokens. **Build domain tools for computation.** A generic shell barely
102 helps open-ended sequence design and analysis (0% to 11%). The next gains lie in biology-specific
103 tool ecosystems and specialist models, not in more conversation between copies of the same model.

104 **Scope.** Our tasks have deterministic answers, and we study short-horizon question answering
105 rather than long-running discovery loops. The frontier and tool comparisons use a 30-question
106 subset, and the tool condition adds a one-sentence note announcing the tools.

107 5 Conclusion

108 On biology research tasks, how a model spends extra inference barely changes which questions it
109 gets right. This holds for voting, critique, debate and longer reasoning alike: the question explains
110 56% of the variance in correctness and the system 0.6%. The ceiling is largely knowledge, and
111 a single tool-using call lifts it where orchestration did not. Life-science agents should spend their
112 budget on tools, not talk.

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A Prompts

All prompts are single user messages. {QUESTION} is the question (with protocol, options or injected files) and {ANSWER INSTRUCTION} is the format line for multiple choice or the official SeqQA2 suffix.

Direct / Independent×3 / Critique call 1 (solve)

Solve the following biology research question.

{QUESTION}

{ANSWER INSTRUCTION}

Respond in exactly this format and keep it short:

<answer>YOUR ANSWER</answer>

CONFIDENCE: <probability between 0 and 1 that your answer is correct>

JUSTIFICATION: <at most two sentences>

195 **Critique call 2 (critique)**

196 Below are a biology research question and a proposed solution. Act as a critical reviewer: identify any
197 scientific, computational or reasoning error in the solution. If you think the answer is wrong, say which
198 answer you believe is correct and why. If you find no error, say so.

199
200 {QUESTION}

201
202 Proposed solution:
203 {CALL 1 OUTPUT}

204
205 Respond with at most four sentences, starting with "CRITIQUE:".

206 **Critique call 3 (revise)**

207 Below are a biology research question, a first attempt at solving it, and a reviewer's critique of that
208 attempt. Either may contain mistakes. Considering both, produce the final answer.

209
210 {QUESTION}

211
212 First attempt:
213 {CALL 1 OUTPUT}

214
215 Reviewer's critique:
216 {CALL 2 OUTPUT}

217
218 {ANSWER INSTRUCTION}

219
220 Respond in exactly this format and keep it short:
221 <answer>YOUR ANSWER</answer>
222 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
223 JUSTIFICATION: <at most two sentences>

224 **Debate call 1 (researcher)**

225 You are a molecular biology researcher. Solve the following biology research question independently.

226
227 {QUESTION}

228
229 {ANSWER INSTRUCTION}

230
231 Respond in exactly this format and keep it short:
232 <answer>YOUR ANSWER</answer>
233 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
234 JUSTIFICATION: <at most two sentences>

235 **Debate call 2 (skeptic)**

236 You are a skeptical scientific reviewer. Solve the following biology research question independently, looking
237 especially for subtle experimental, biological or computational errors and traps.

238
239 {QUESTION}

240
241 {ANSWER INSTRUCTION}

242
243 Respond in exactly this format and keep it short:
244 <answer>YOUR ANSWER</answer>
245 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
246 JUSTIFICATION: <at most two sentences>

247 **Debate call 3 (adjudicator)**

248 Two scientists independently analysed the biology research question below. Adjudicate between their analyses
249 (either or both may be wrong) and submit the final answer.

250
251 {QUESTION}

252

253 Analysis 1 (molecular biology researcher):
 254 {CALL 1 OUTPUT}
 255
 256 Analysis 2 (skeptical scientific reviewer):
 257 {CALL 2 OUTPUT}
 258
 259 {ANSWER INSTRUCTION}
 260
 261 Respond in exactly this format and keep it short:
 262 <answer>YOUR ANSWER</answer>
 263 CONFIDENCE: <probability between 0 and 1 that your answer is correct>
 264 JUSTIFICATION: <at most two sentences>

265 B Pre-registration deviations

266 All task IDs, prompts, decoding settings and the primary analysis were frozen after a 10-question
 267 pilot and before any main-set call. Changes made after the freeze: (i) throttle and retry fixes for an
 268 undocumented 150 requests/hour provider limit (no completed call was re-run); (ii) 22 GPT-OSS
 269 questions served in full by a second provider of the same model (identical prompts, seeds and cap;
 270 excluding them leaves DEBATE – VOTE at -1.6), with partially served questions discarded and re-
 271 run on the primary provider; (iii) the reasoning-effort and tool arms on the fixed 30-question subset
 272 and the unsolved set, added after the freeze. Frontier models were served through their vendor CLI
 273 with web search, code execution and sub-agents disabled except in the tool condition (a first attempt
 274 with web search left on was discarded unanalysed); GPT-OSS at high effort was served by a second
 275 host whose low-effort accuracy matches the main run. The tool condition prepends one sentence
 276 announcing the tools, and all searches and commands were logged and checked for benchmark
 277 access (none).

278 C Additional results

Category	Gemma					Lite					OSS				
	D	I	C	M	Disagree	D	I	C	M	Disagree	D	I	C	M	Disagree
DbQA	56.7	53.3	40.0	56.7	33.3	40.0	40.0	40.0	33.3	26.7	46.7	46.7	43.3	36.7	36.7
LitQA2	33.3	30.0	33.3	36.7	46.7	60.0	53.3	53.3	46.7	26.7	20.0	30.0	40.0	36.7	43.3
ProtocolQA	56.7	56.7	60.0	70.0	20.0	66.7	80.0	86.7	73.3	26.7	56.7	50.0	43.3	46.7	33.3
SeqQA	56.7	56.7	66.7	73.3	46.7	73.3	73.3	53.3	66.7	13.3	63.3	66.7	43.3	60.0	33.3
SeqQA2	40.0	46.7	36.7	50.0	63.3	66.7	66.7	46.7	66.7	40.0	46.7	53.3	43.3	56.7	43.3

Table 2: Per-category accuracy (D=Direct, I=Independent×3, C=Critique, M=Multi). Numbers are percentages.